

Cube Values

MA.8.AR.2.3

I can evaluate a cube root and find every real solution of $x^3 = q$.

Yesterday: two answers, one answer, or none.

Today: always exactly one. The reason is the count of negative factors.

Warm-Up

On your own. Four minutes.

Same and different

A square with area 64 has a side of 8. A cube with volume 64 has an edge of 4.

1. What is the same about the two situations?
2. What is different?
3. Why do the same 64 give two different answers?

Notes

I. Cubing means multiplying a number by itself three times

A. **cube (a number)** — to multiply the number by itself three times

B. **4 cubed** = 4^3 = $4 \times 4 \times 4$ = **64**.

C. **It is NOT 4×3 . The small 3 counts appearances, the same as the small 2 did.**

D. The volume of a cube is $V = s \times s \times s = s^3$.

Notes

II. The perfect cubes, and their negatives

- A. **perfect cube** — a number you get by cubing an integer
- B. **1, 8, 27, 64, 125**
- C. **and their negatives: $-1, -8, -27, -64, -125$**
- D. Five numbers and their opposites. Worth as many points as yesterday's longer list.

Notes

III. A negative number has a cube root

- A. **The index is written this time. $\sqrt[3]{q}$ means the cube root of q .**
- B. **$\sqrt[3]{27} = 3$, because $3 \times 3 \times 3 = 27$.**
- C. **$\sqrt[3]{-27} = -3$, because $(-3) \times (-3) \times (-3) = -27$.**
- D. **Three negative factors give a NEGATIVE product. Two would have given a positive.**

Notes

IV. Solving $x^3 = q$ — always exactly one

If q is ...	How many solutions	Example
POSITIVE	one, and it is positive	$x^3 = 8$ gives $x = 2$
ZERO	one, and it is 0	$x^3 = 0$ gives $x = 0$
NEGATIVE	one, and it is negative	$x^3 = -8$ gives $x = -2$

Example 1

Two cube roots. Answer out loud.

$$\sqrt[3]{125}$$

$$\sqrt[3]{-64}$$

Which of the two will have a negative answer, and how do you know before you compute it?

Example 1

Both, worked.

$$\sqrt[3]{125} = 5 \quad \text{because } 5 \times 5 \times 5 = 125$$

$$\sqrt[3]{-64} = -4 \quad \text{because } (-4) \times (-4) \times (-4) = -64$$

Example 1

Why the second one is negative.

$$(-4) \times (-4) \times (-4)$$

the first two cancel the third one stays

Example 2

Solve it. Whiteboards.

$$x^3 = -27$$

How many solutions, and what are they?

Example 2

The solution.

$$\mathbf{x = \sqrt[3]{-27} = -3}$$

$$\text{Check: } (-3) \times (-3) \times (-3) = -27 \quad \checkmark$$

Example 2

And why there is no second one.

$$\text{Try } x = 3: \quad 3 \times 3 \times 3 = 27$$

27 is not -27. ✗

Example 3

Same 64. One solution or two?

$$x^2 = 64$$

$$x^3 = 64$$

Take a vote before anybody writes anything.

Example 3

Try both signs on each.

$\mathbf{x^2 = 64}$	$\mathbf{x^3 = 64}$
$8 \times 8 = 64 \quad \checkmark$	$4 \times 4 \times 4 = 64 \quad \checkmark$
$(-8) \times (-8) = 64 \quad \checkmark$	$(-4) \times (-4) \times (-4) = -64 \quad \times$

Example 3

So the counts differ.

$x^2 = 64$	$x^3 = 64$
TWO solutions, $x = \pm 8$	ONE solution, $x = 4$

Whiteboards • Question 1 of 9

Boards up on three.

$$\sqrt[3]{64}$$

Answer it.

A number, or a sentence. Read the index first.

Whiteboards • Question 1 of 9

Answer.

$$\sqrt[3]{64}$$

$$4 \times 4 \times 4 = 64$$

Answer: 4

Whiteboards • Question 2 of 9

Boards up on three.

$$\sqrt[3]{-64}$$

Answer it.

A number, or a sentence. Read the index first.

Whiteboards • Question 2 of 9

Answer.

$$\sqrt[3]{-64}$$

$$(-4) \times (-4) \times (-4) = -64$$

Answer: -4

Whiteboards • Question 3 of 9

Boards up on three.

$$\sqrt{64}$$

Answer it.

A number, or a sentence. Read the index first.

Whiteboards • Question 3 of 9

Answer.

$$\sqrt{64}$$

$$8 \times 8 = 64$$

Answer: 8

Whiteboards • Question 4 of 9

Boards up on three.

$$\sqrt{-64}$$

Answer it.

A number, or a sentence. Read the index first.

Whiteboards • Question 4 of 9

Answer.

$$\sqrt{-64}$$

nothing real squares to a negative

Answer: no real value

Whiteboards • Question 5 of 9

Boards up on three.

$$x^3 = 64$$

Answer it.

A number, or a sentence. Read the index first.

Whiteboards • Question 5 of 9

Answer.

$$x^3 = 64$$

only 4 works

Answer: $x = 4$

Whiteboards • Question 6 of 9

Boards up on three.

$$x^2 = 64$$

Answer it.

A number, or a sentence. Read the index first.

Whiteboards • Question 6 of 9

Answer.

$$x^2 = 64$$

8 and -8 both work

Answer: $x = \pm 8$

Whiteboards • Question 7 of 9

Boards up on three.

A cube-shaped box holds 125 cubic inches. How long is one edge?

Give the edge, with its unit.

A number, or a sentence. Read the index first.

Whiteboards • Question 7 of 9

Answer.

A cube-shaped box holds 125 cubic inches. How long is one edge?

$$5 \times 5 \times 5 = 125$$

Answer: 5 inches

Whiteboards • Question 8 of 9

Boards up on three.

$$x^3 = -27$$

Answer it.

A number, or a sentence. Read the index first.

Whiteboards • Question 8 of 9

Answer.

$$x^3 = -27$$

$$(-3) \times (-3) \times (-3) = -27$$

Answer: $x = -3$

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Priya says $\sqrt[3]{-64}$ has no real value, because you cannot take the root of a negative number. Is she right?

Write your answer in sentences.

This one is written work. Say when her rule IS true.

Whiteboards • Question 9 of 9

Answer.

Priya says $\sqrt[3]{-64}$ has no real value, because you cannot take the root of a negative number. Is she right?

$$(-4) \times (-4) \times (-4) = -64$$

Answer: No — $\sqrt[3]{-64} = -4$

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Whatever you do not finish is tonight's practice.
4. The skills for today are posted in Google Classroom.